TRACE

Tracking, Recognition, Analytics & City-wide Traffic Enforcement

Technical Requirement Document — Revised v1.2

Changes from v1.1: NFR-05 is made operational through a P2 M2 failure-isolation checklist. The single-FastAPI architecture and API contract remain unchanged.

1.2 Architectural Principle — UNCHANGED FROM v1.1

TRACE uses five loosely-coupled logical modules corresponding to the Layer Model. Each has a defined input/output contract. For the prototype, these modules run inside one FastAPI application rather than separate deployable microservices.

2. System Architecture — UNCHANGED FROM v1.1

Camera Feed → Perception Module → Identity Fusion Module → Road-Graph Store → Spatial-Temporal Reasoning Module → [Trajectory | Analytics | Anomaly & Prediction] → Alert Module → FastAPI API Layer → React/MapLibre Dashboard.

3.1 Prototype Stack — UNCHANGED

Purpose	Technology
Video ingestion	Recorded/simulated multi-camera feeds via OpenCV
Detection	YOLO (v8/v9 family)
Tracking	ByteTrack
OCR	PaddleOCR
Identity fusion	Python weighted scoring
Road graph	PostgreSQL + PostGIS
Backend/API	FastAPI (Python), one application containing logical modules
Realtime	WebSockets (polling fallback)
Frontend	React + MapLibre GL + charting library
Deployment	Docker Compose

5. API Design Overview — UNCHANGED

The existing FastAPI API contract is retained exactly; no endpoint is added, removed, renamed or otherwise changed.

Endpoint	Method	Purpose
/vehicles/{plate}/trajectory	GET	Reconstructed chronological trajectory.
/analytics/heatmap	GET	Current density values per segment.
/analytics/od-matrix	GET	Current OD matrix.
/analytics/segments/{id}	GET	Average speed, density and congestion status.
/analytics/forecast/{segment_id}	GET	Short-horizon forecast.
/alerts	GET	Recent blacklist/anomaly alerts.
/blacklist	POST / GET	Add/read watched plates.
/ws/live-updates	WS	Live observation, alert and heatmap updates.

7. NFR-05 Reliability — OPERATIONAL CHECKLIST FOR P2 / M2

NFR-05 requires graceful degradation when a camera feed is unavailable. The following checklist is the implementation-level sign-off for M2. It applies to the single-FastAPI prototype and does not change the external API.

- 1. Wrap each camera's perception call in its own try/except. A bad frame, corrupted feed or OCR crash must log and skip that camera's cycle; it must never raise past the ingestion loop.
- 2. Never let a per-camera failure block the shared event loop. Run perception/ingestion for each camera as an independent async task or background job so one hung feed cannot stall detection on the others.
- 3. On failure, write the camera's status as degraded or down, using the existing cameras table status field, instead of leaving stale data. This status is consumed by the dashboard and Camera Network Status view.
- 4. Set a per-camera timeout on detection/OCR calls so a slow frame cannot hang that camera's pipeline indefinitely.
- 5. Keep the FastAPI process itself unaffected by any single module exception. A Layer 3 reasoning bug on one trajectory query must return a clean error response rather than crash requests for other endpoints.
- 6. M2 manual test: kill or corrupt one simulated camera feed mid-run. Confirm the other cameras continue ingesting normally and the dashboard shows the affected camera as down rather than frozen or blank.

M2 Sign-off Evidence

P2 records the result of the six checks in the M2 QA notes, including the camera used for the failure test and evidence that unaffected cameras continued processing. P3 validates the UI state and P1 verifies the corresponding dashboard/camera-status presentation.

8. NFR-08 Maintainability — UNCHANGED FROM v1.1

The five intelligence layers remain separable code modules with defined interfaces inside one FastAPI application. A layer can be modified or replaced without rewriting the others.

All other TRD sections and the API contract remain unchanged.